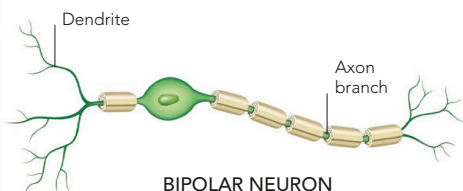


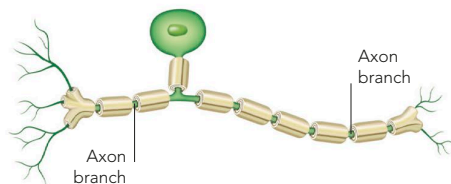
TYPES OF NEURON

Neuron cell bodies vary greatly in size and shape, as do the type, number, and length of their projections. Neurons can be unipolar, bipolar, or multipolar. Unipolar neurons are found mainly in the sensory nerves of the PNS. Bipolar neurons exist mostly in the embryo, but adults have some in the retina of the eye and the olfactory nerve in the nose. Most neurons in the brain and spinal cord are multipolar.



BIPOLAR NEURON

The nerve cell body is located between two processes – an axon and a dendrite.



UNIPOLAR NEURON

A single short process, an axon, extends from the nerve cell body and splits into two.



MULTIPOLAR NEURON

These nerve cell bodies have three or more processes – one axon and several dendrites.

NERVES

Nerves, which resemble rope-like cords, pass between and branch into the body's organs and tissues. They are composed of bundles of communication strands – the elongated axons or nerve fibres of neurons. Each bundle is known as a fascicle. Most nerves carry two types of fibre. Sensory, or afferent, fibres bring messages from receptors in the sense organs and other structures to the spinal cord and brain. Motor, or efferent, fibres convey signals from the brain or spinal cord to a muscle or gland. Some nerves contain just sensory fibres, such as the optic nerve, while others have solely motor fibres.

INSIDE A NERVE

Bundles of nerve fibres are embedded within tough connective tissue to protect them from damage.

